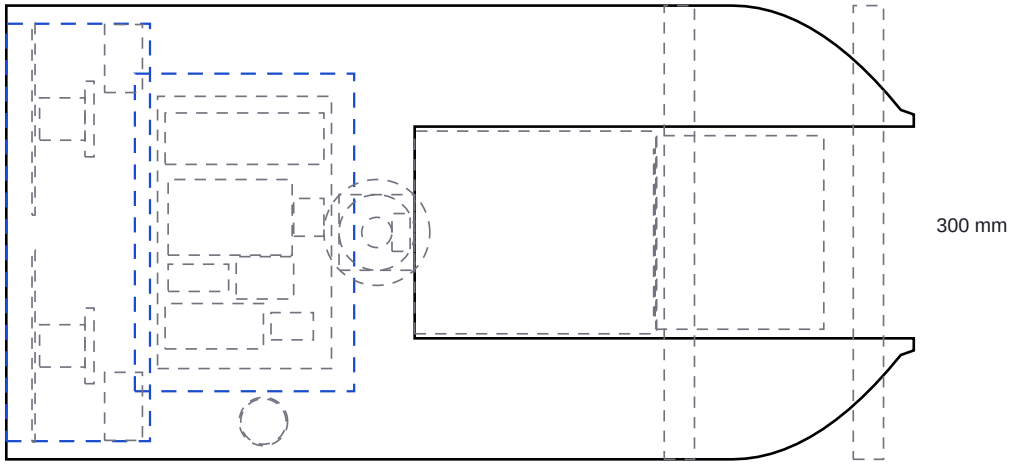
<- must measure 50 mm

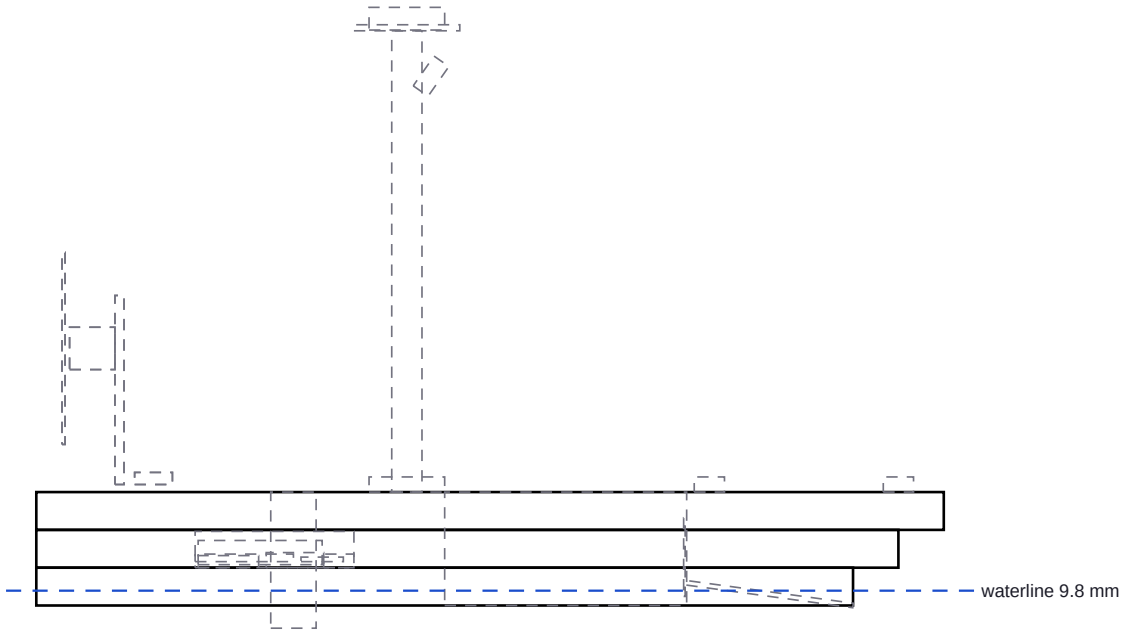
VARUNA U-Hull Block

600 x 300 mm thermocol boat | Easiest to build

PLAN (from above), scale 1:5



SIDE, scale 1:5



Size	600 x 300 mm, hull 75 mm tall, 396 mm to GPS top
Weight	1144 g empty, 1451 g with 307 g of plastic
Draft	9.8 mm empty, 12.4 mm full
Reserve buoyancy	6.7 kg before the deck goes under
Stability (GM)	833 mm: very stable
Trim	bow 6.2 mm higher than stern empty; slide the battery to level
Collection	1.54 L basket
Camera	340 mm above water, tilted 35 deg down
Battery	LiPo 3S 2200 mAh (24 Wh), about 0.8 h
Thermocol	3 sheets 1000 x 500, or 1 sheet 2 x 3 ft (610 x 915)

How to use these templates

1. Print at 100% / Actual size. Check the 50 mm bar with a ruler.
2. Join tiles by matching the grey 50 mm grid and the tile letters.
3. Stick the paper on thermocol with a glue stick.
4. Cut with a hot-wire cutter or a new snap-off blade, held upright.
5. Stack layers on skewers through the + holes. Glue with Fevicol SH.
6. Never use Fevikwik, spray paint or solvent glue: they melt thermocol.
7. Sand the bow steps smooth, coat with PVA + tissue or packing tape, then water-based acrylic paint.
8. Float-test in a tub before fitting the electronics.

- cut line
- cut out (window / pocket)
- fold / glue line
- placement guide, do not cut
- skewer hole: lines up the layers

VARUNA U-Hull Block | Hull layer 1 (bottom) - tile A1 of B2 (25 mm thermocol)

<- must measure 50 mm

A1

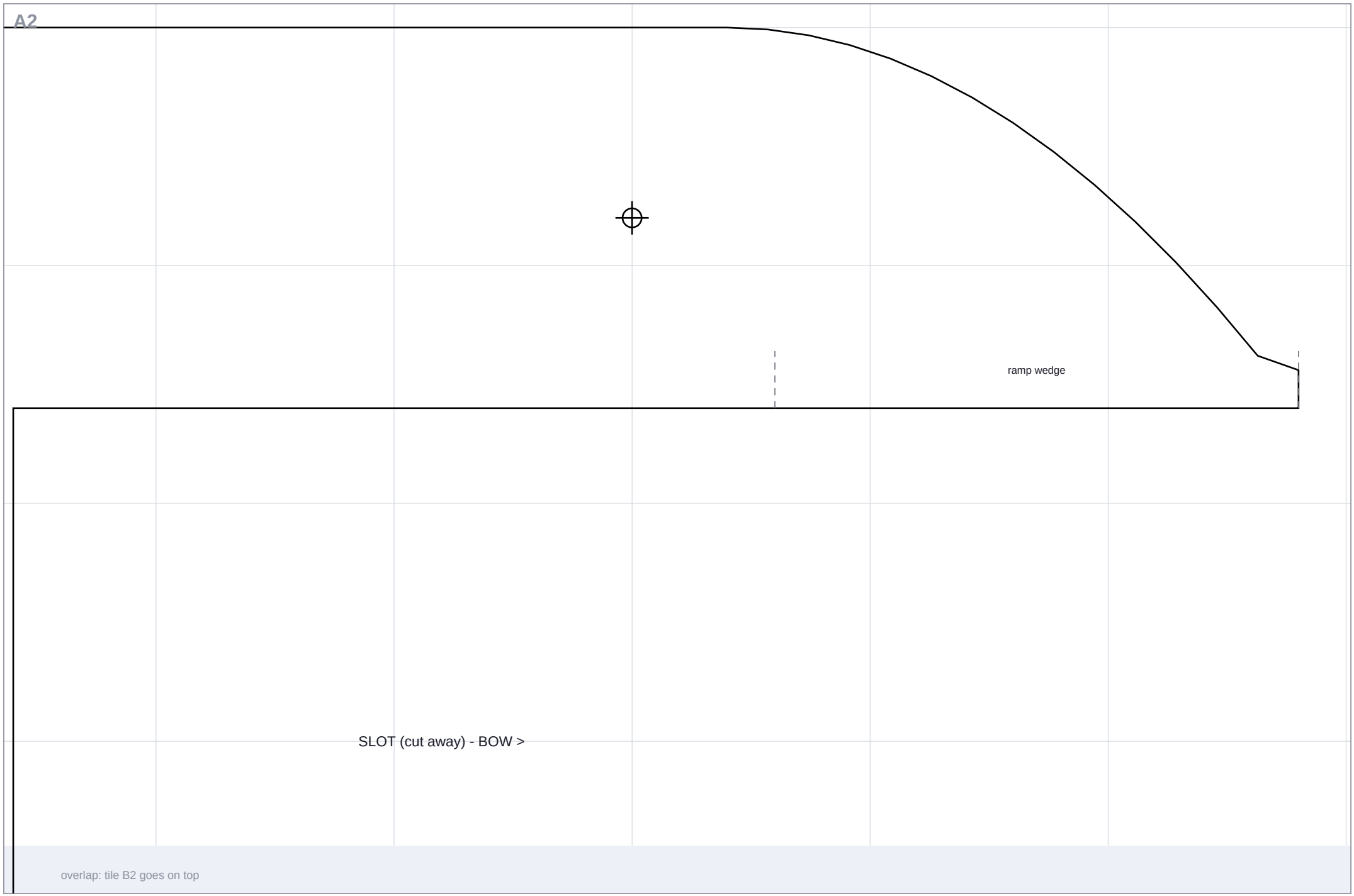
HULL layer 1 (bottom)

overlap: tile A2 goes on top

overlap: tile B1 goes on top

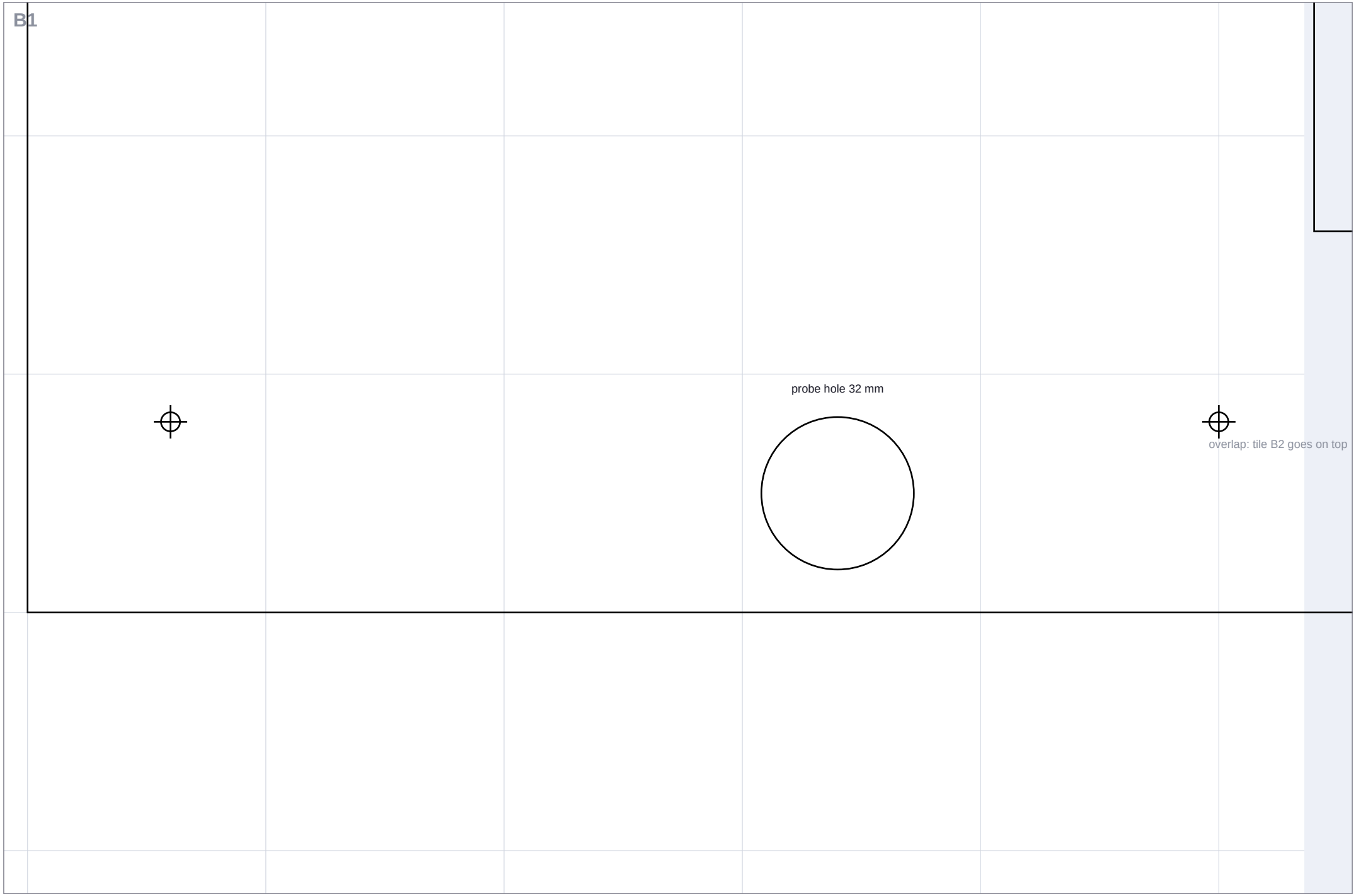
VARUNA U-Hull Block | Hull layer 1 (bottom) - tile A2 of B2 (25 mm thermocol)

<- must measure 50 mm



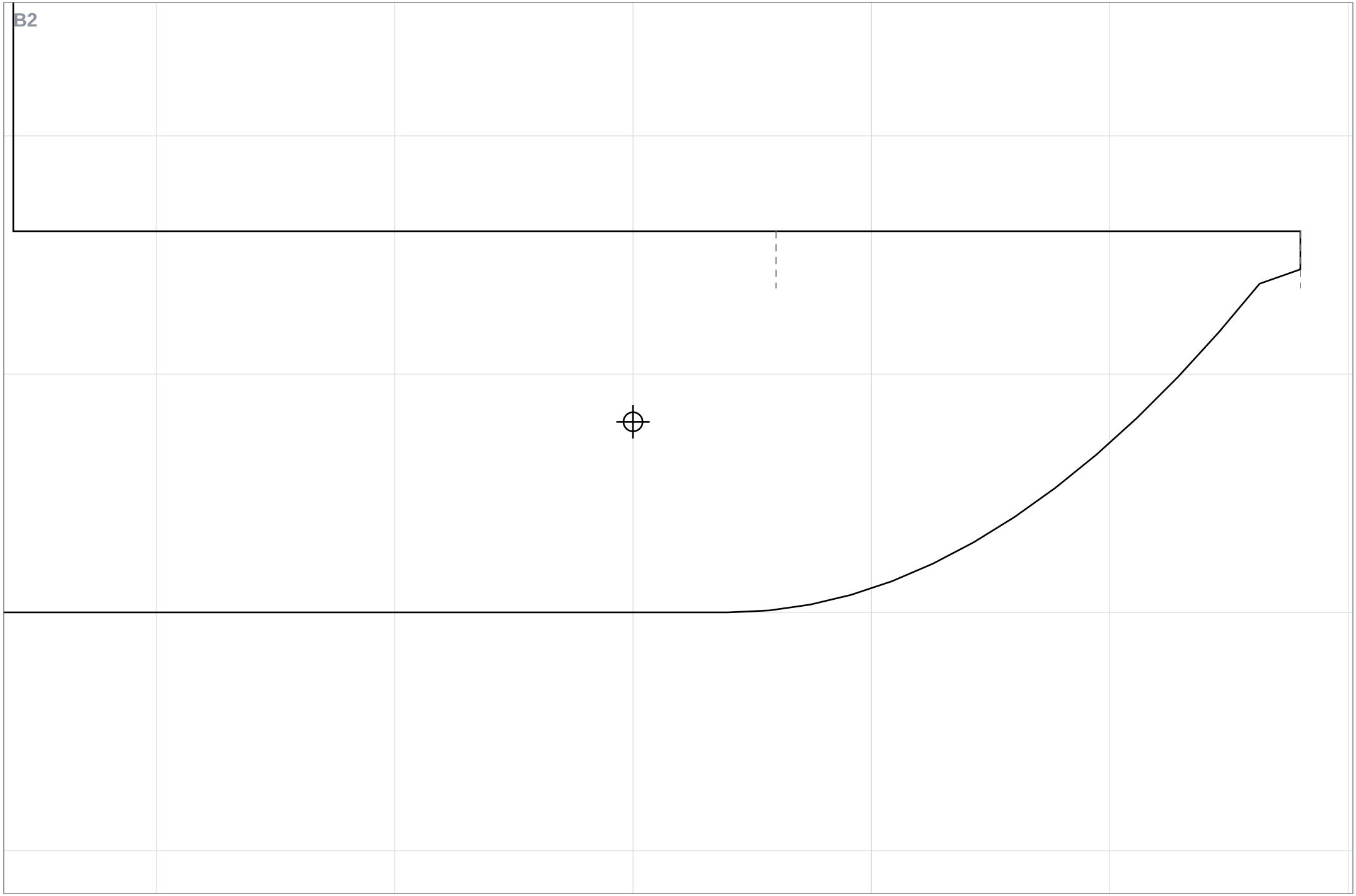
VARUNA U-Hull Block | Hull layer 1 (bottom) - tile B1 of B2 (25 mm thermocol)

<- must measure 50 mm



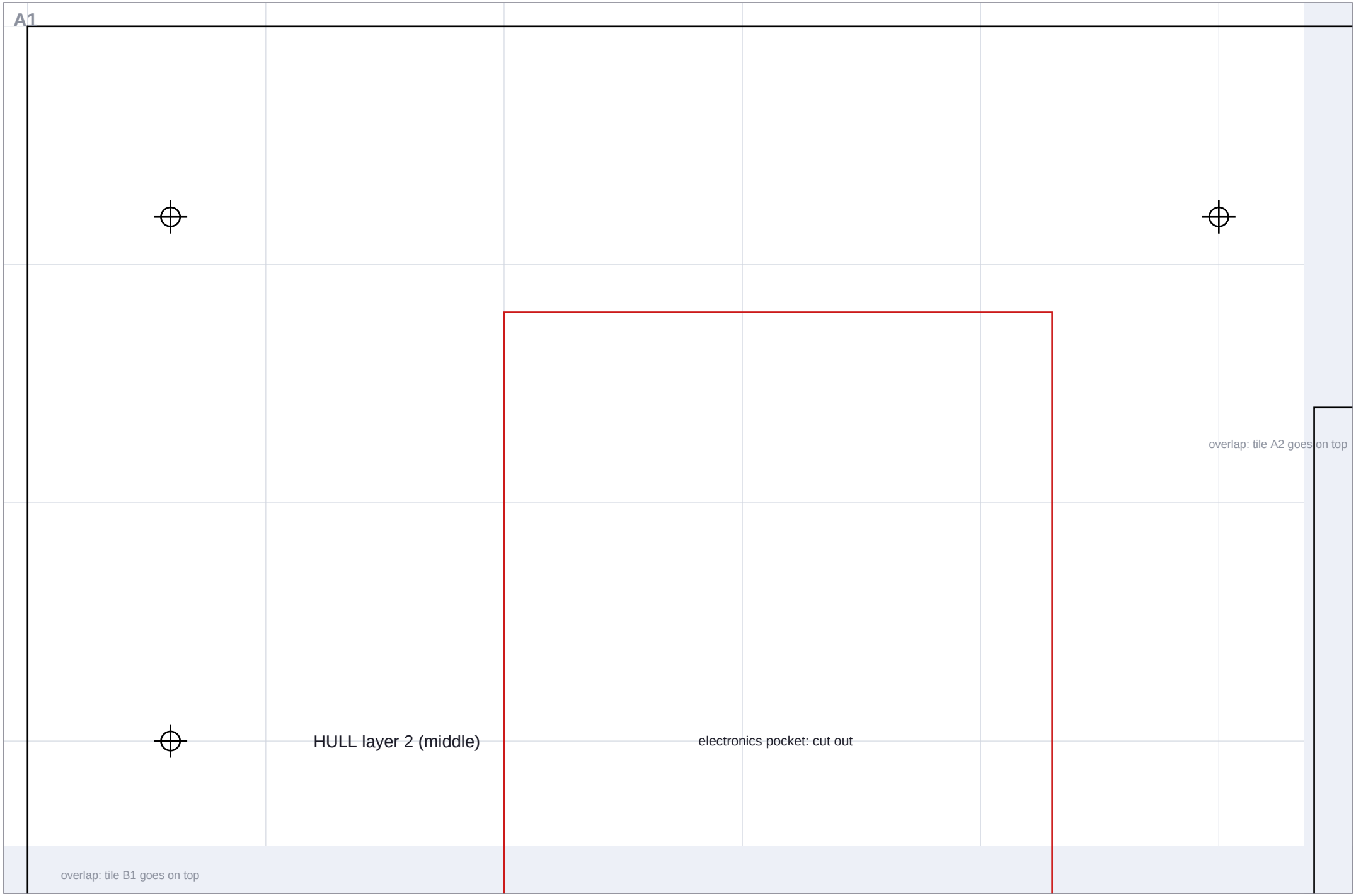
VARUNA U-Hull Block | Hull layer 1 (bottom) - tile B2 of B2 (25 mm thermocol)

<- must measure 50 mm

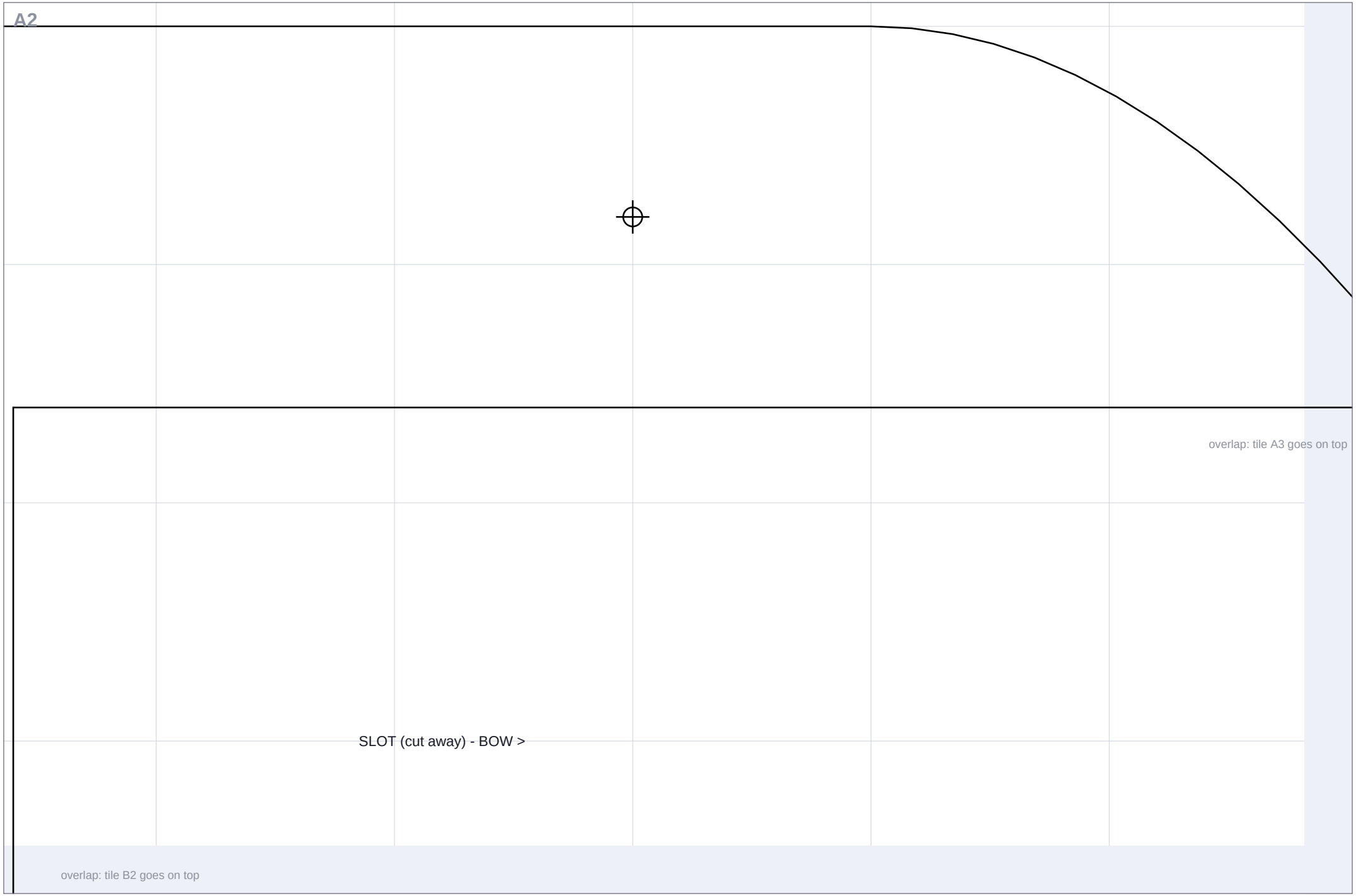


VARUNA U-Hull Block | Hull layer 2 (middle) - tile A1 of B3 (25 mm thermocol)

<- must measure 50 mm

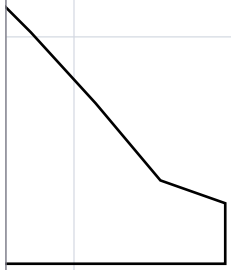


<- must measure 50 mm



<- must measure 50 mm

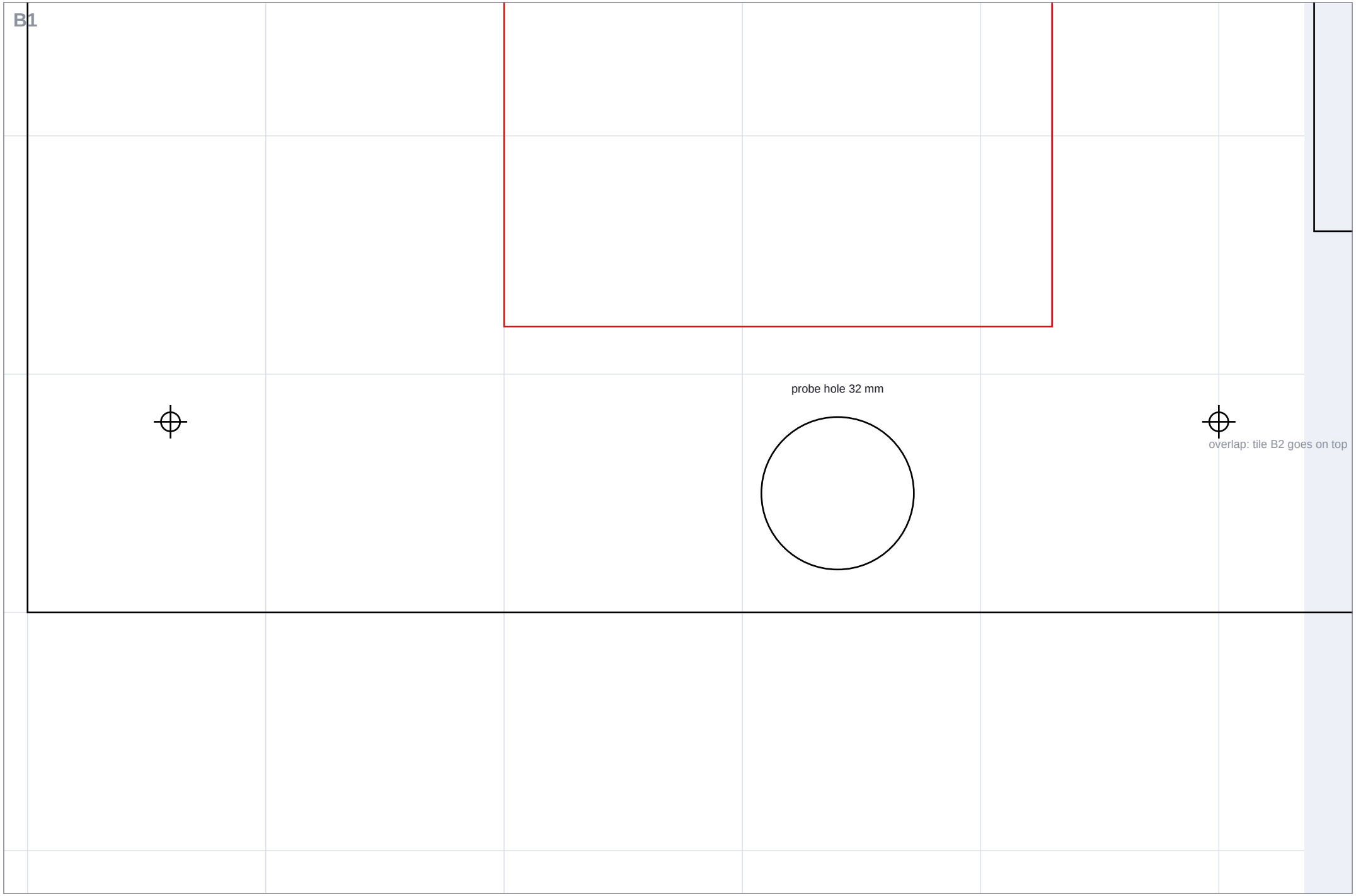
A3



overlap: tile B3 goes on top

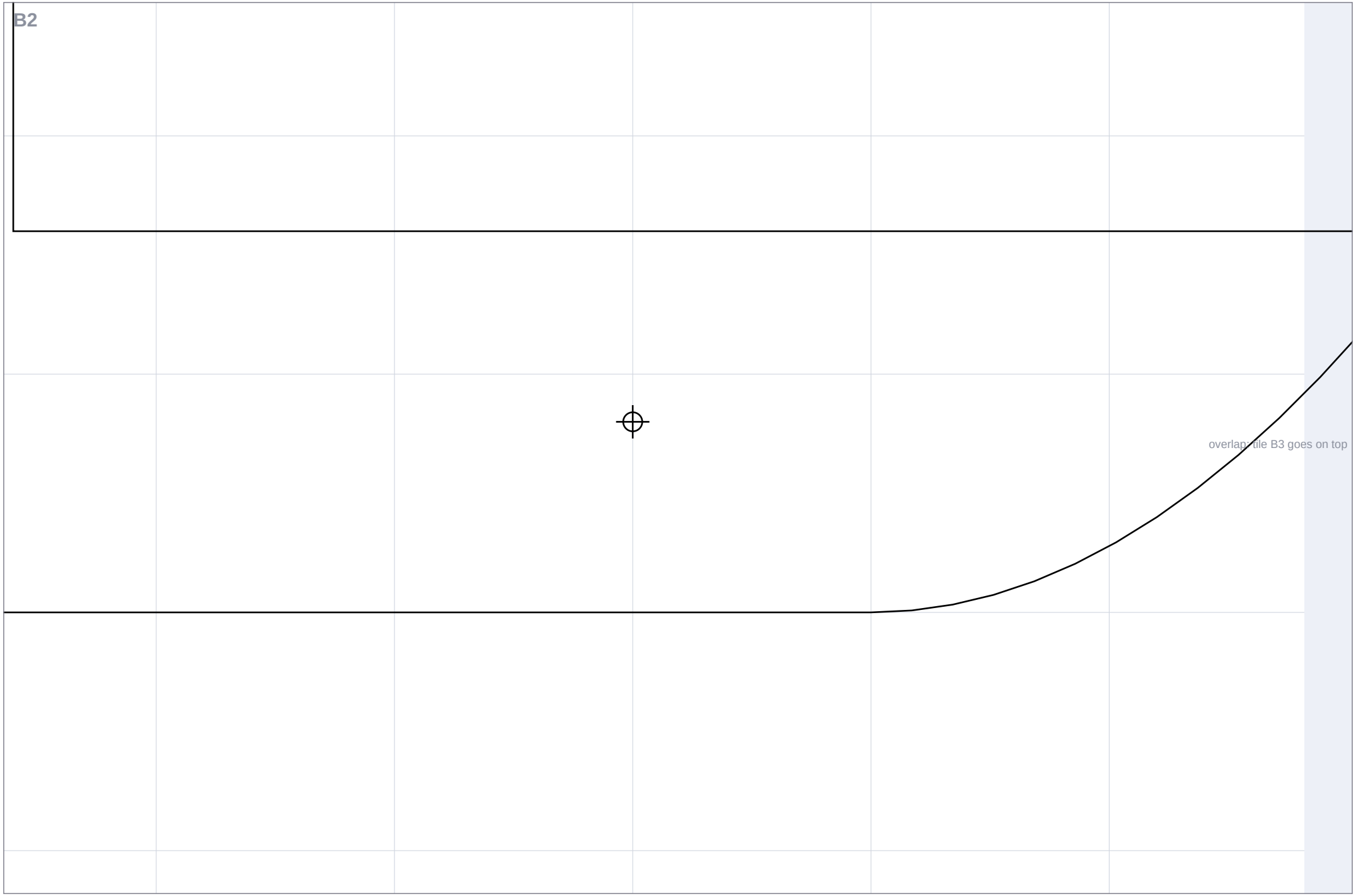
VARUNA U-Hull Block | Hull layer 2 (middle) - tile B1 of B3 (25 mm thermocol)

<- must measure 50 mm



VARUNA U-Hull Block | Hull layer 2 (middle) - tile B2 of B3 (25 mm thermocol)

<- must measure 50 mm

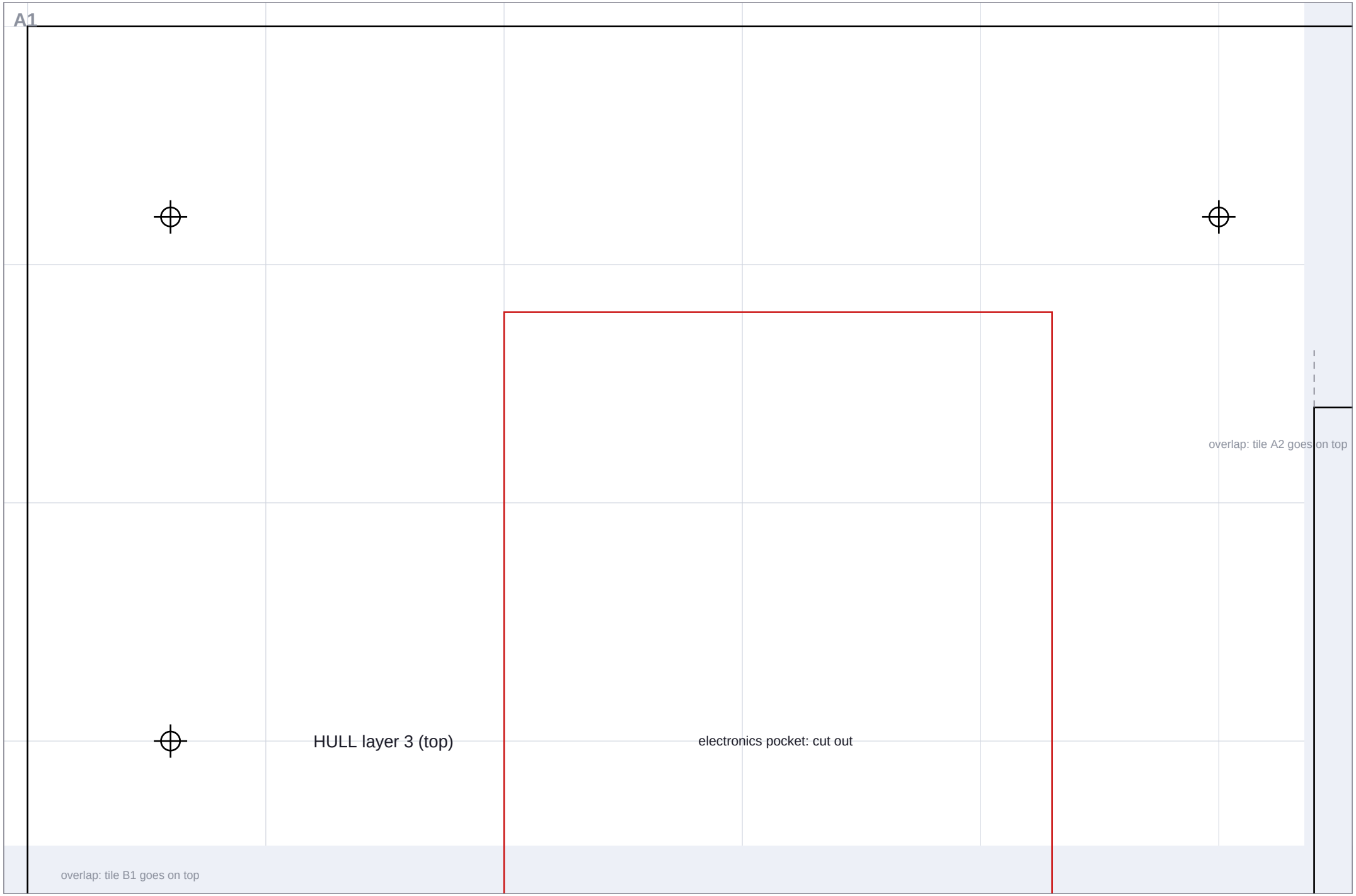


<- must measure 50 mm

B3

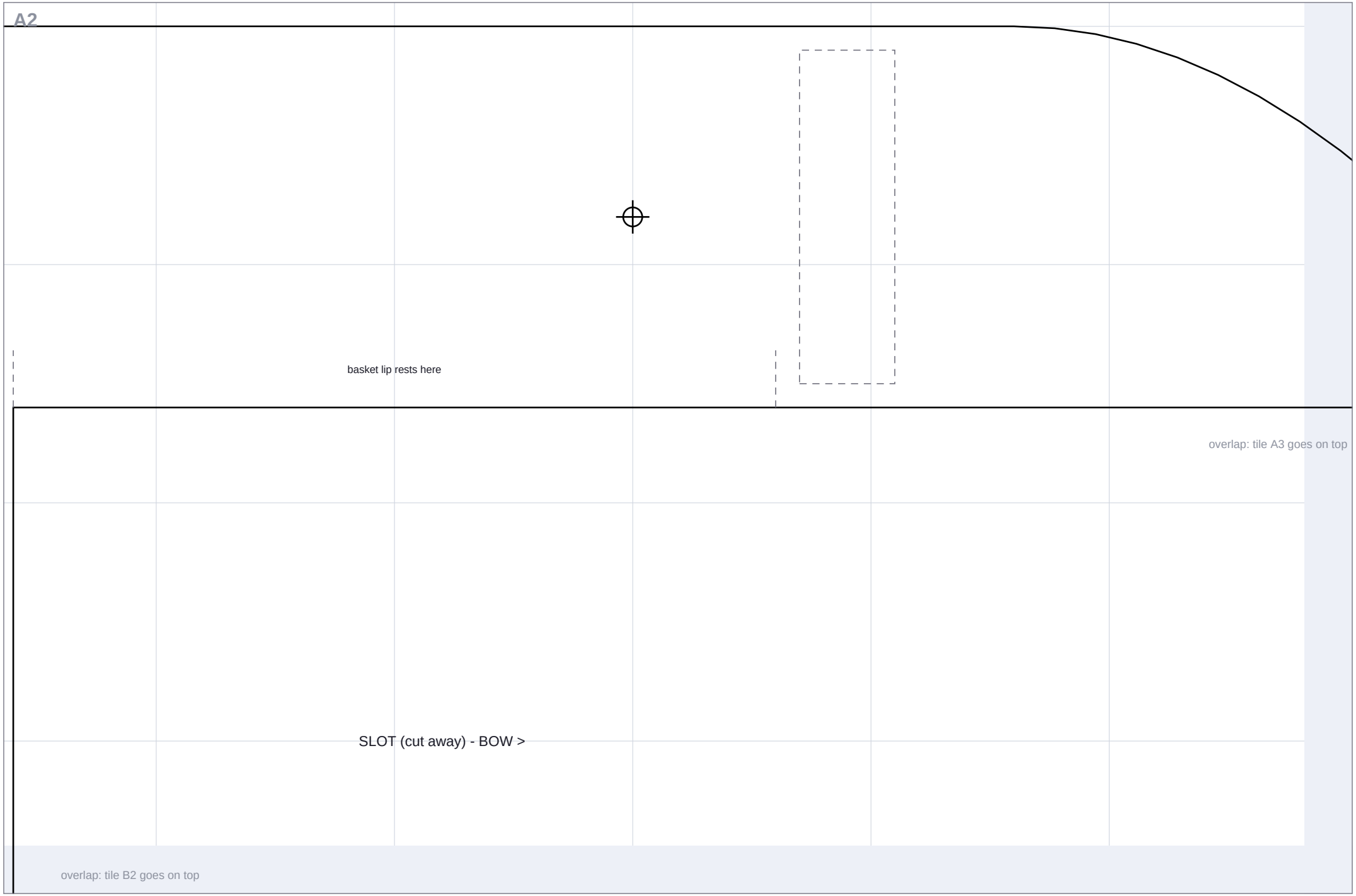
VARUNA U-Hull Block | Hull layer 3 (top) - tile A1 of B3 (25 mm thermocol)

<- must measure 50 mm



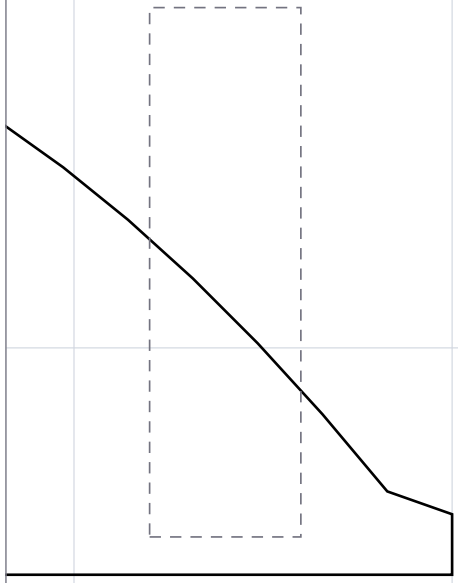
VARUNA U-Hull Block | Hull layer 3 (top) - tile A2 of B3 (25 mm thermocol)

<- must measure 50 mm



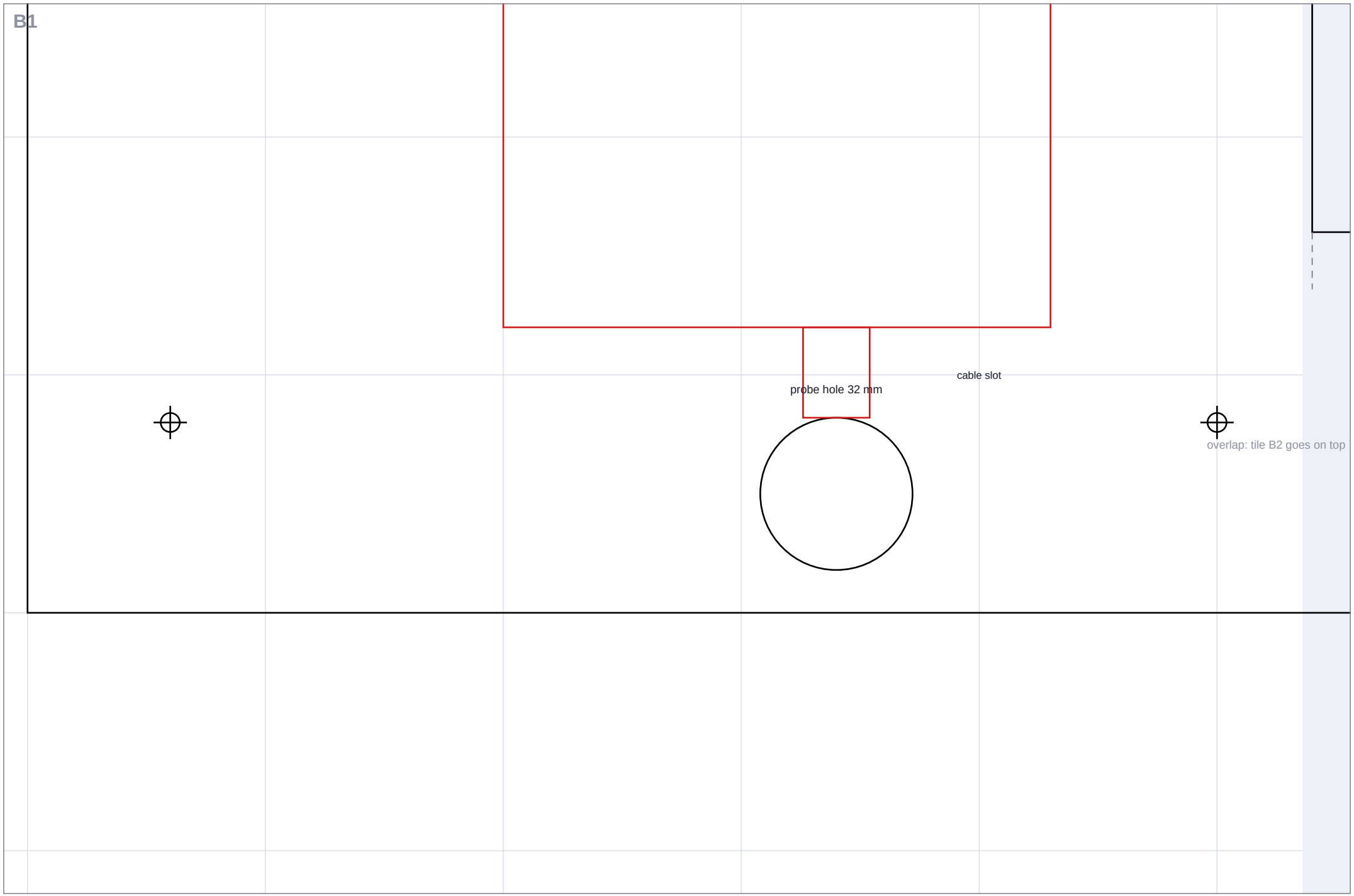
<- must measure 50 mm

A3



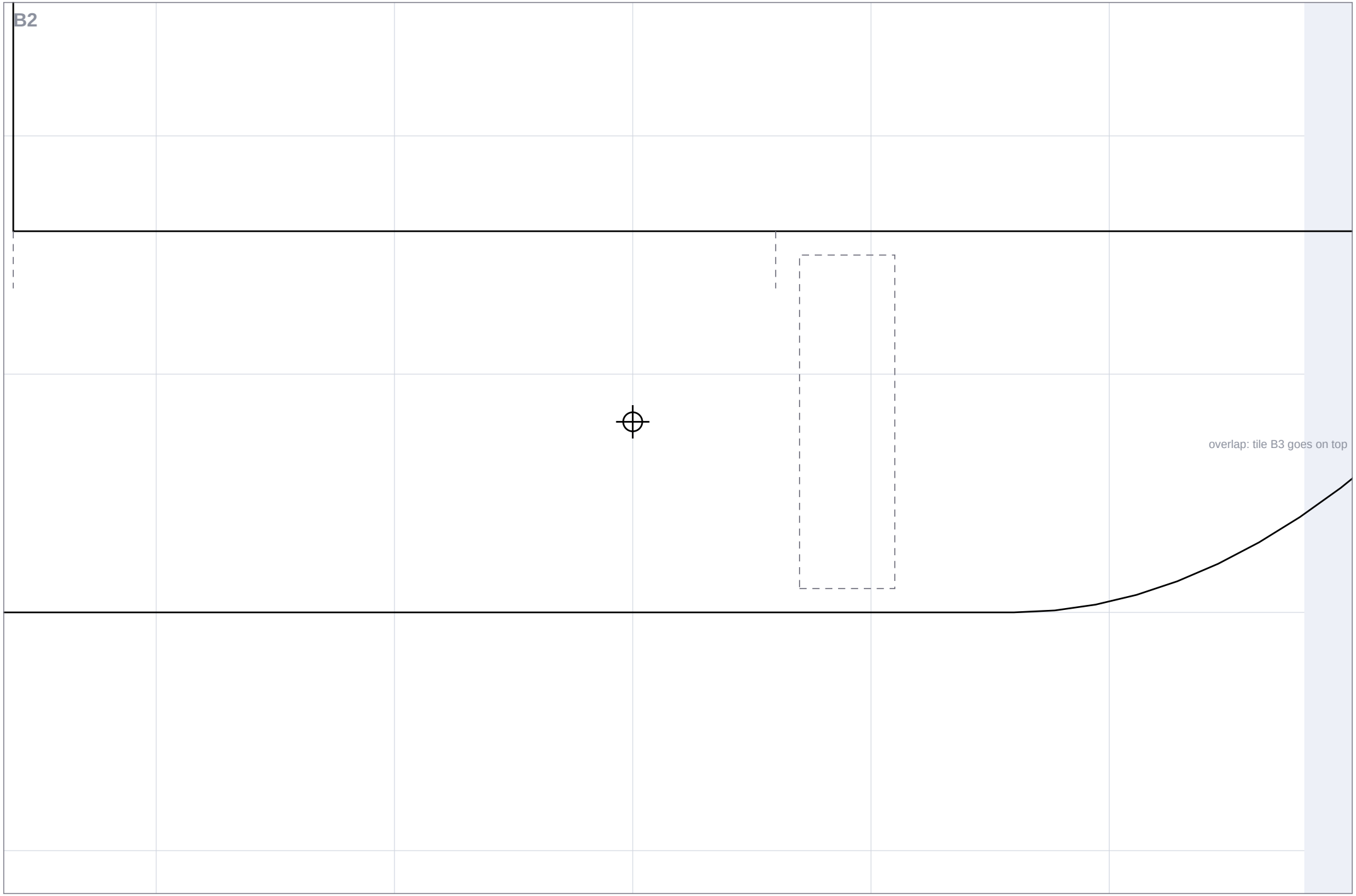
overlap: tile B3 goes on top

VARUNA U-Hull Block | Hull layer 3 (top) - tile B1 of B3 (25 mm thermocol)



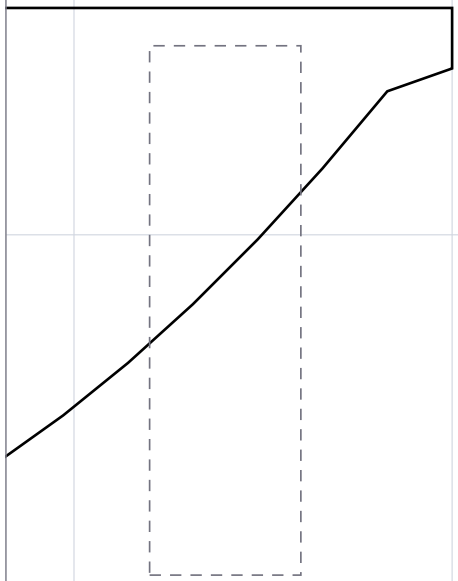
VARUNA U-Hull Block | Hull layer 3 (top) - tile B2 of B3 (25 mm thermocol)

<- must measure 50 mm



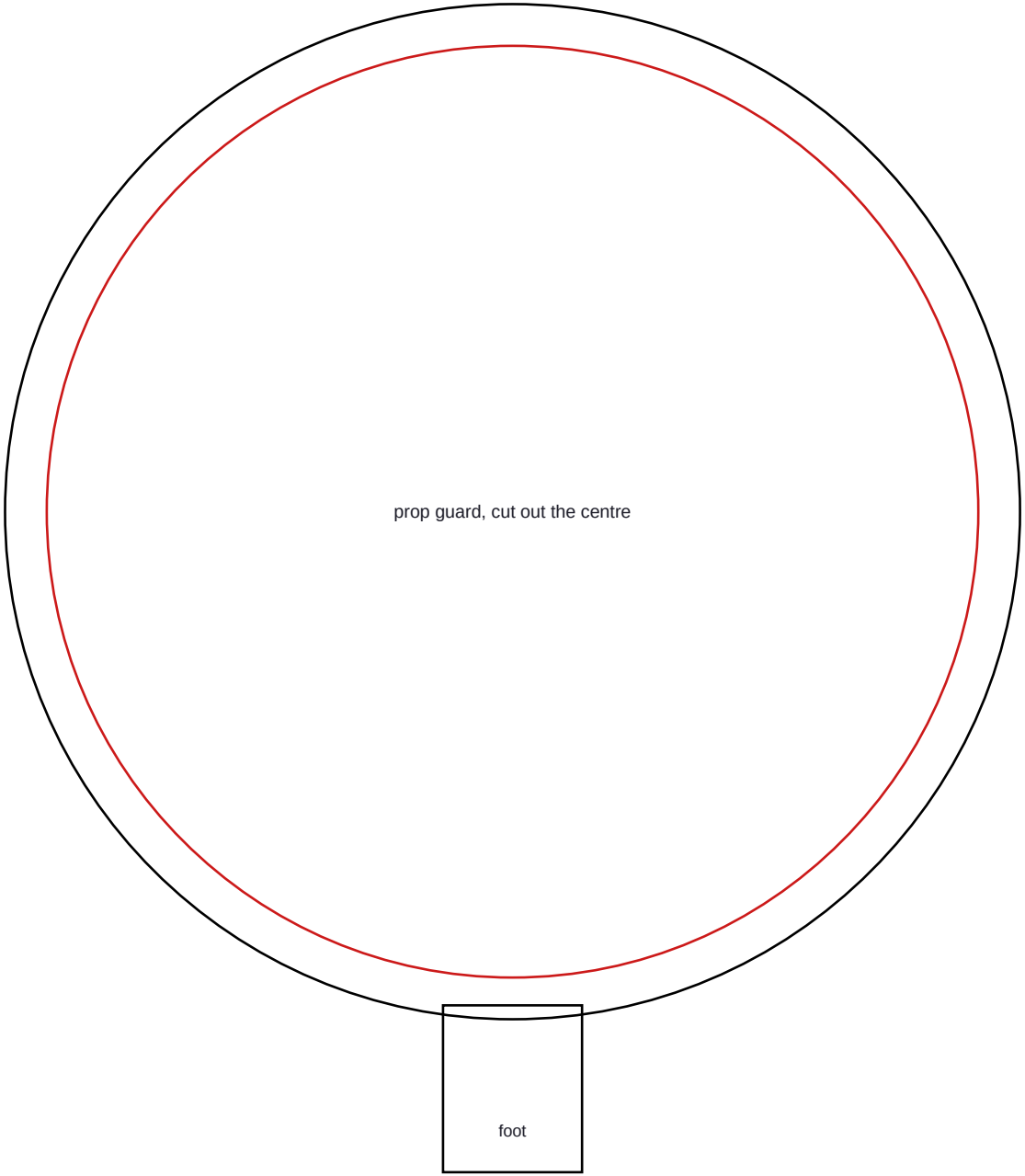
<- must measure 50 mm

B3



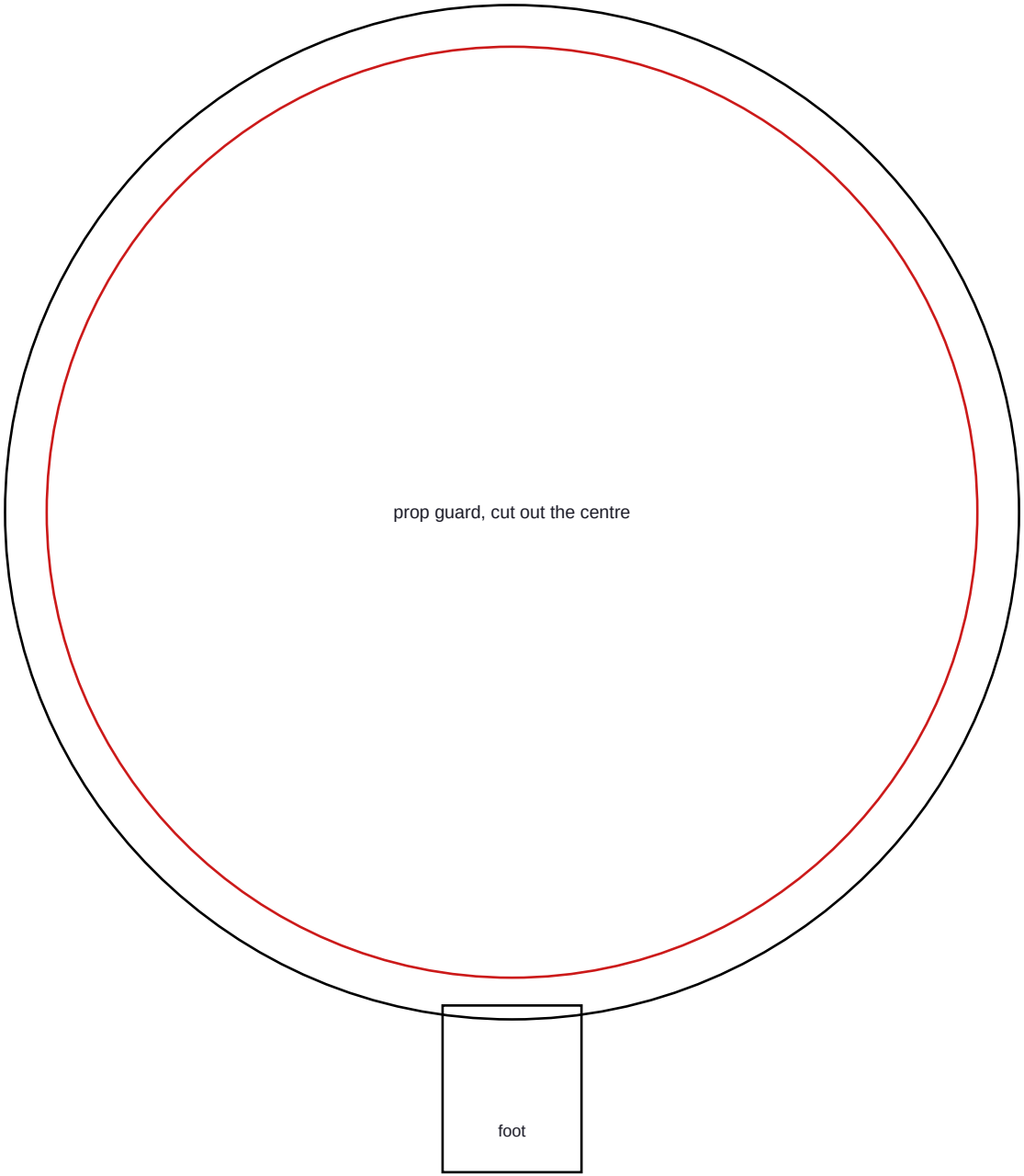
<- must measure 50 mm

Prop guard ring (left)
5 mm sunboard



<- must measure 50 mm

Prop guard ring (right)
5 mm sunboard



<- must measure 50 mm

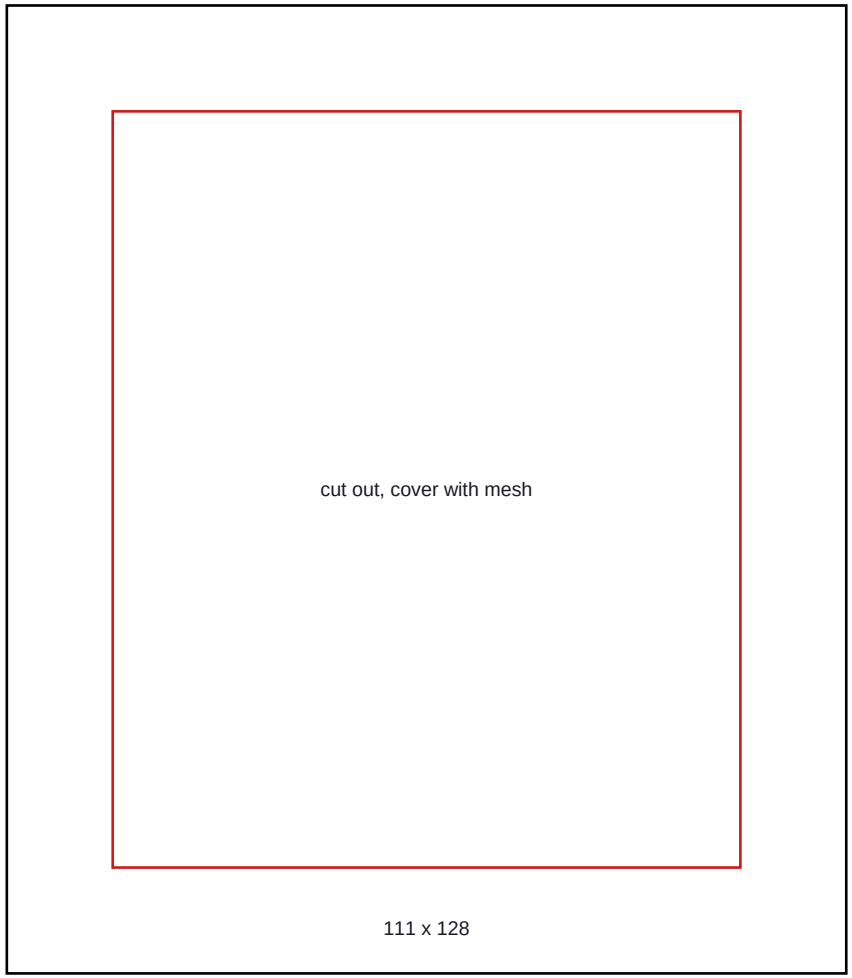
Hatch lid

5 mm sunboard

<- must measure 50 mm

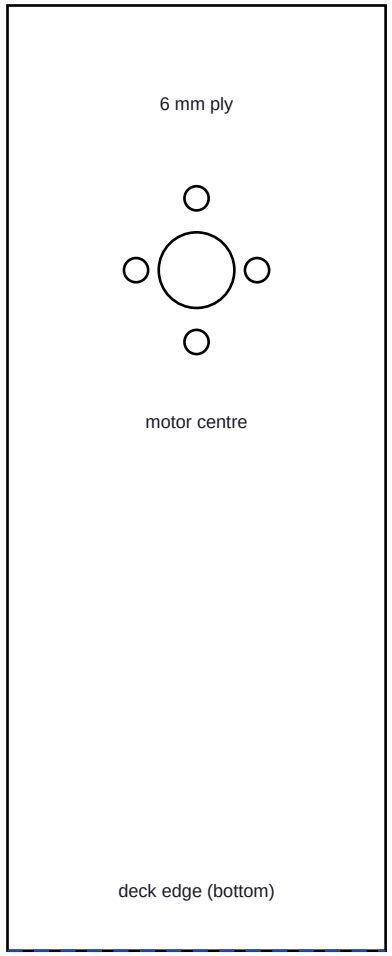
Basket bottom
3 mm sunboard

Intake ramp frame
3 mm sunboard

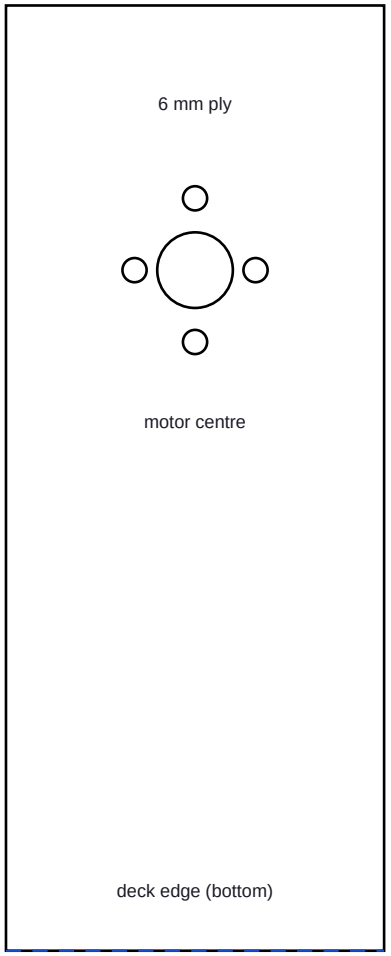


<- must measure 50 mm

Motor tower plate (left)
6 mm plywood (or 2 x 5 mm sunboard)



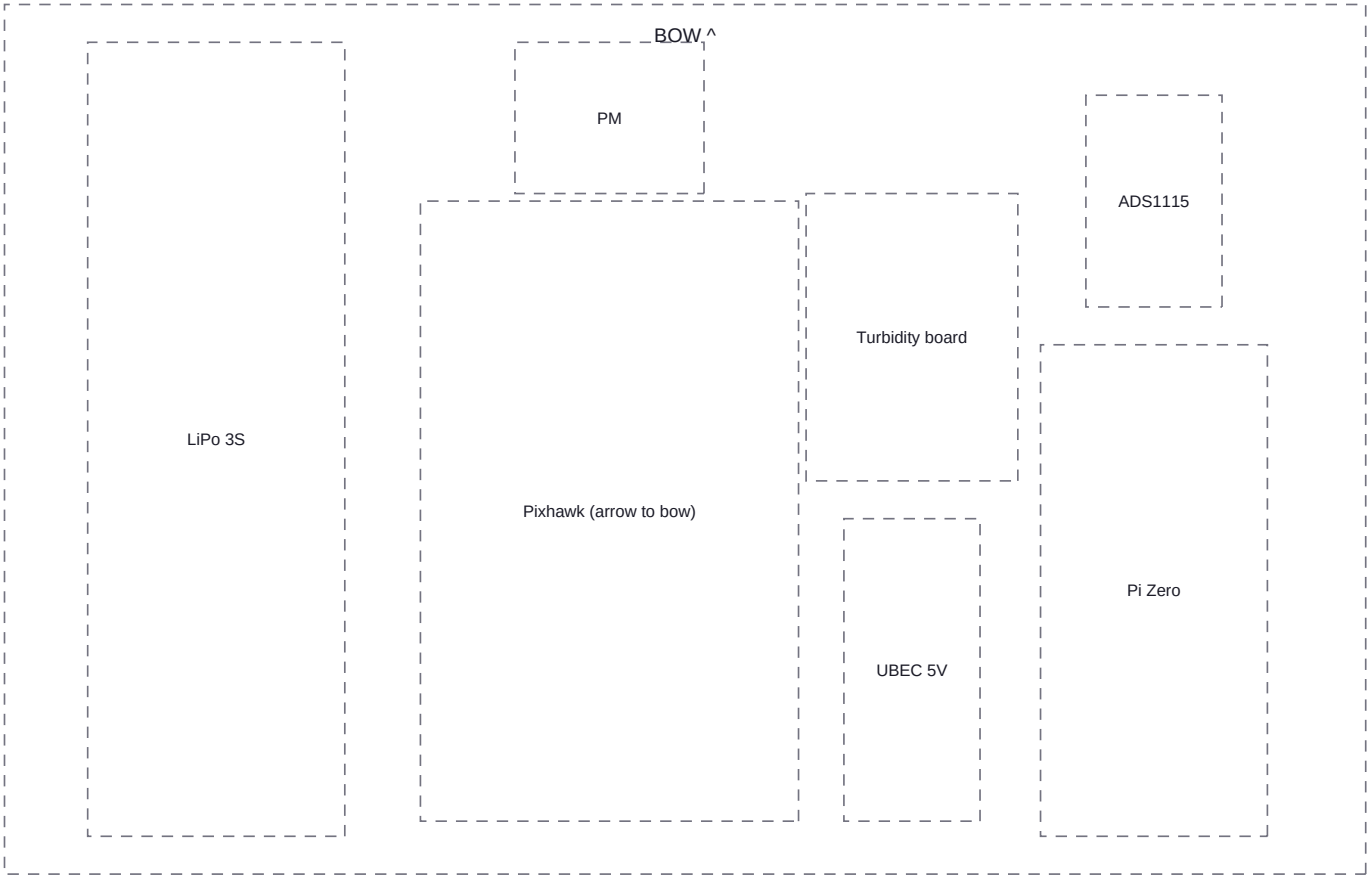
Motor tower plate (right)
6 mm plywood (or 2 x 5 mm sunboard)



<- must measure 50 mm

Pocket floor guide

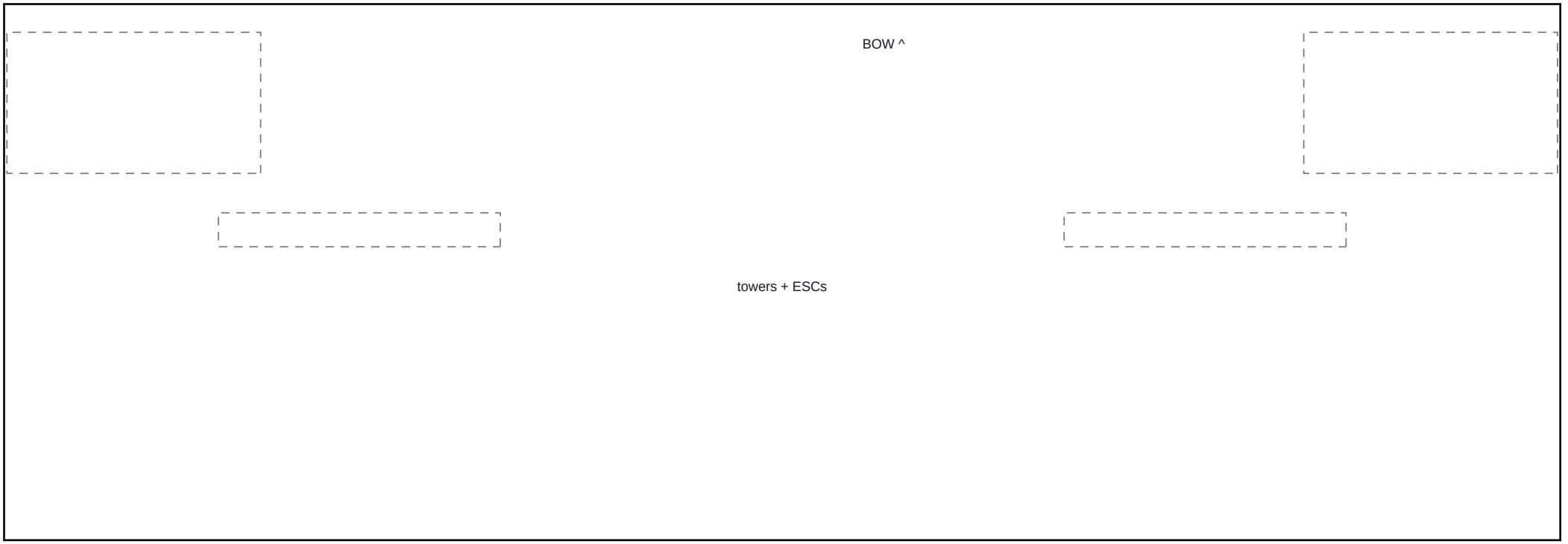
paper guide, glue inside the floor



<- must measure 50 mm

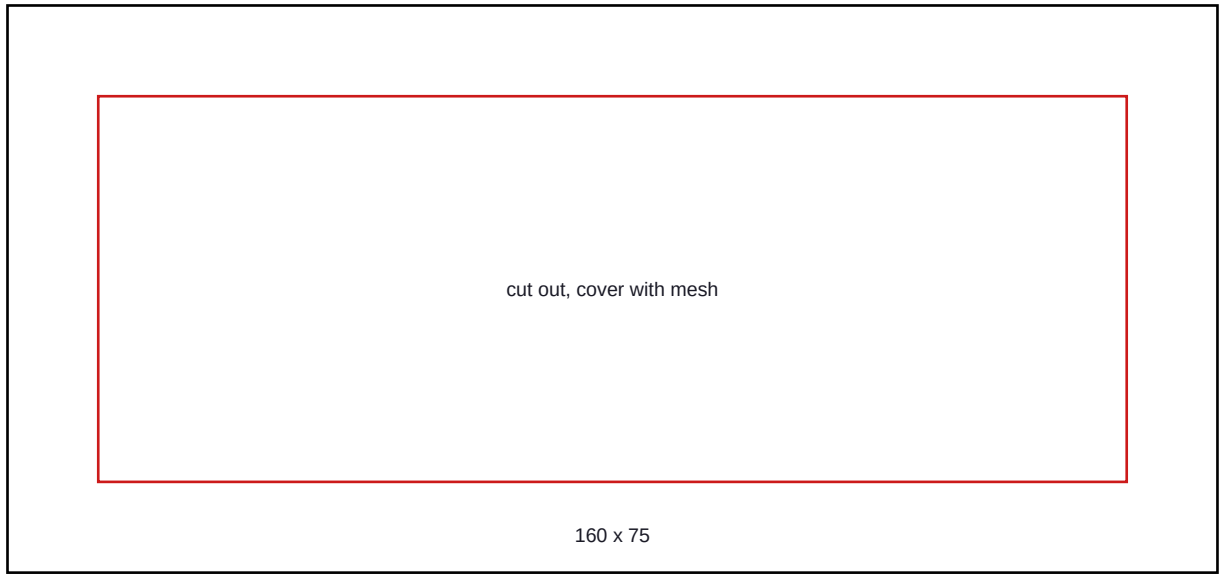
Stern plate

5 mm sunboard

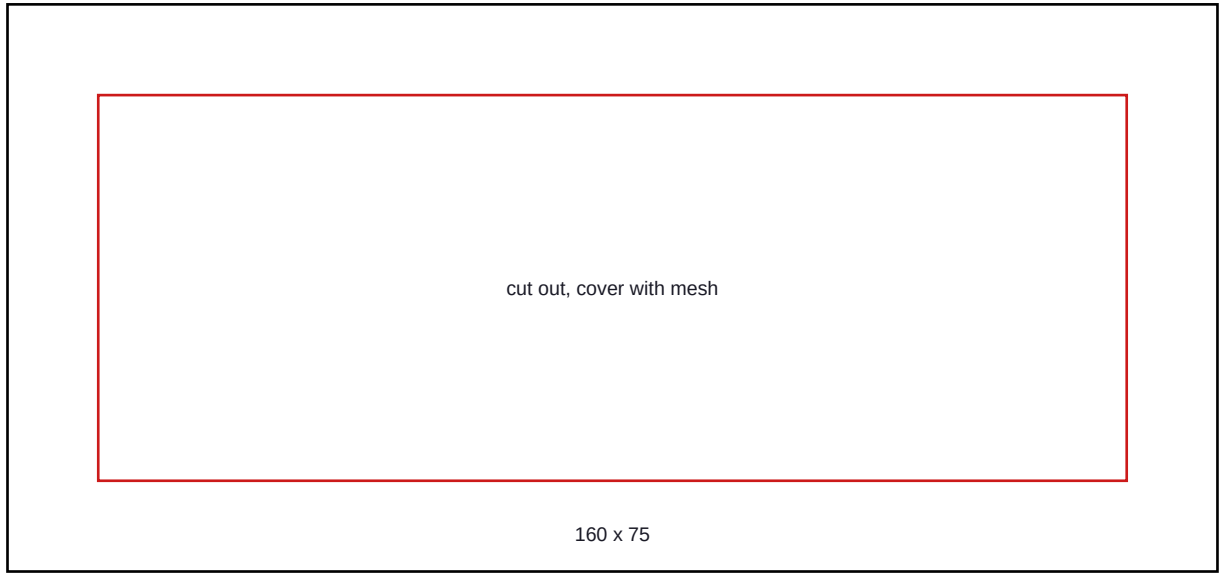


<- must measure 50 mm

Basket side (left)
3 mm sunboard

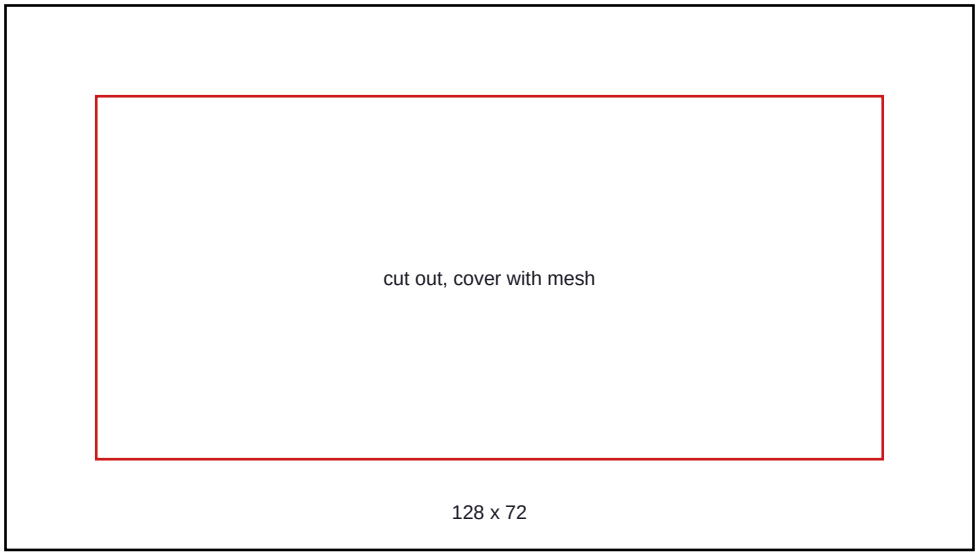


Basket side (right)
3 mm sunboard

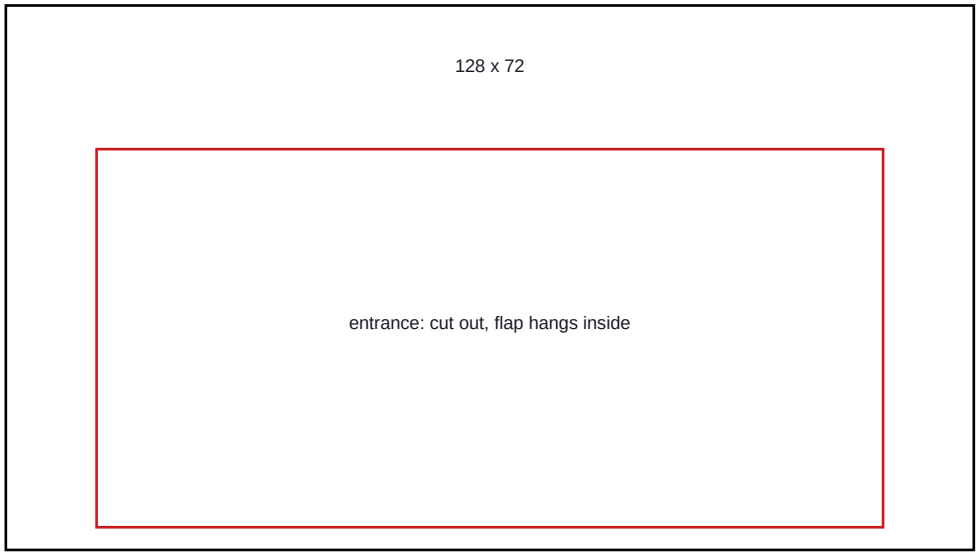


<- must measure 50 mm

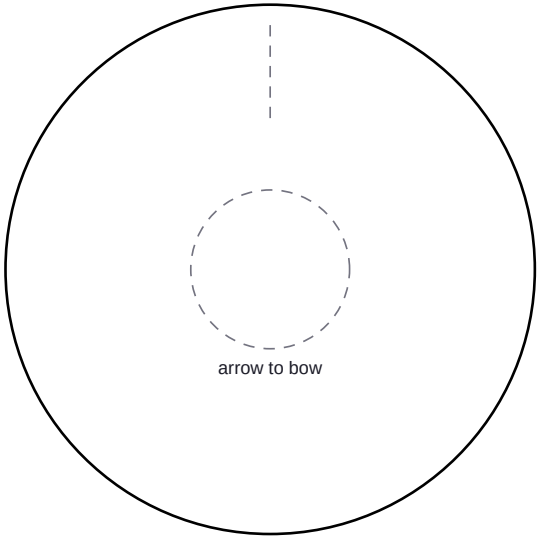
Basket back end
3 mm sunboard



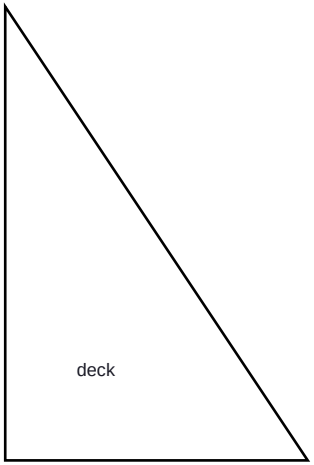
Basket front end (entrance)
3 mm sunboard



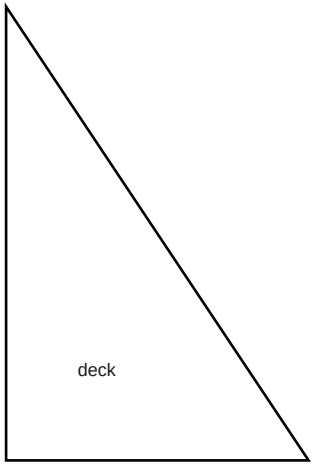
GPS disc
5 mm sunboard



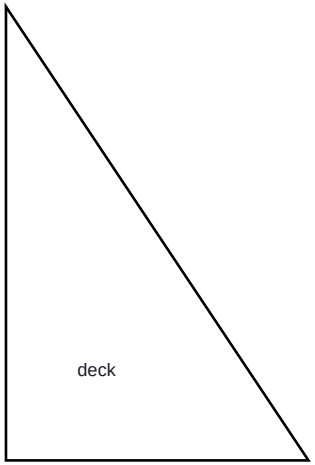
Tower gusset
5 mm sunboard



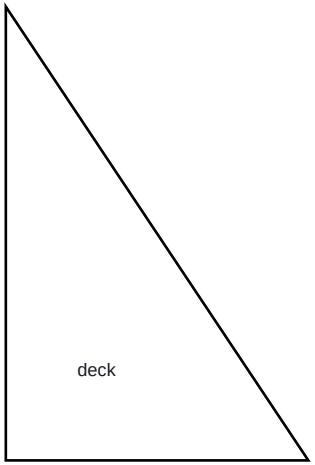
Tower gusset
5 mm sunboard

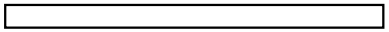


Tower gusset
5 mm sunboard



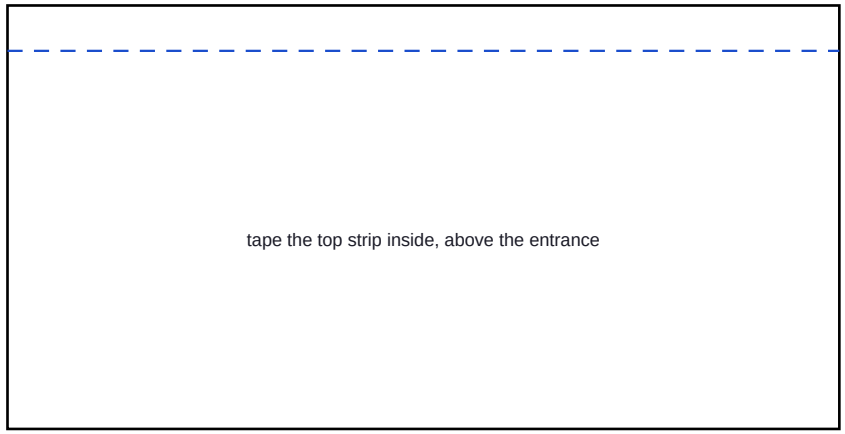
Tower gusset
5 mm sunboard



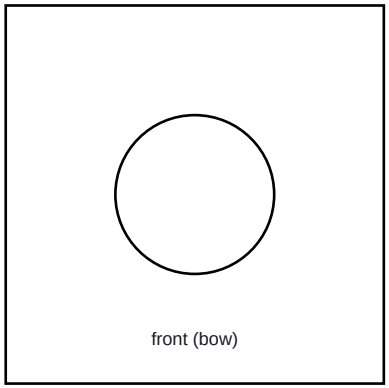


<- must measure 50 mm

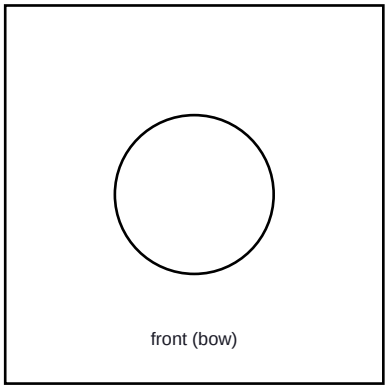
One-way flap
PET sheet from a bottle / transparency



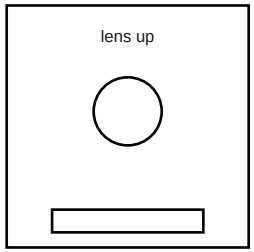
Mast socket layer
5 mm sunboard (cut 2, top one with hole)



Mast socket layer
5 mm sunboard (cut 2, top one with hole)



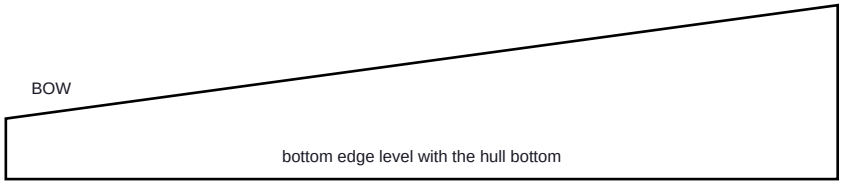
Camera plate
3 mm sunboard



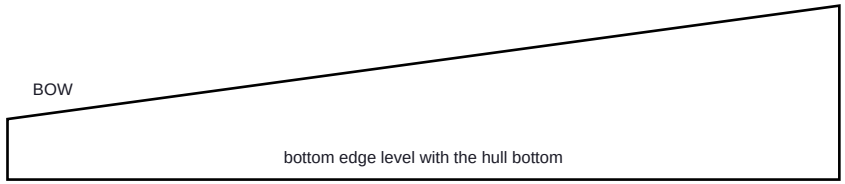
Basket lip (left)
3 mm sunboard

Basket lip (right)
3 mm sunboard

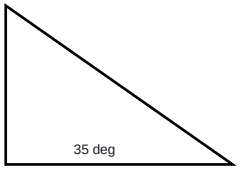
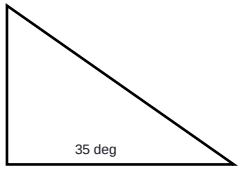
Ramp support wedge
5 mm sunboard (cut 2)



Ramp support wedge
5 mm sunboard (cut 2)



Camera wedge side (35 deg)
3 mm sunboard



Camera wedge side (35 deg)
3 mm sunboard